
Empirical Evidence for Alignment Faking in a Small LLM and Prompt-Based Mitigation Techniques

Jeanice Koorndijk
Seraphion Technology

Abstract

Current literature suggests that alignment faking is an emergent property of large language models. We present the first empirical evidence that a small instruction-tuned model, specifically LLaMA 3 8B, can also exhibit alignment faking. We further show that prompt-only interventions, including deontological moral framing and scratchpad reasoning, significantly reduce this behavior without modifying model internals. This challenges the assumption that prompt-based interventions are trivial and that deceptive alignment requires scale. We introduce a taxonomy distinguishing shallow deception, shaped by context and suppressible through prompting, from deep deception, which reflects persistent, goal-driven misalignment. Our findings refine the understanding of deception in language models and underscore the need for deceptive alignment evaluations across model sizes and deployment settings.

1 Introduction

Alignment faking is a type of alignment failure in which a model appears aligned during training or evaluation, while internally pursuing objectives that differ from those intended by its developers, evaluators, or users (inner misalignment). This can occur even when the training/evaluation metric is well-chosen and accurately reflects the developers' real aim (outer alignment) [Hubinger et al., 2019, Greenblatt et al., 2024]. This type of deceptive alignment (where deception is a misleading appearance) poses a safety risk: once outer alignment no longer benefits the model, such as when oversight is removed during deployment, it may act in unexpected or unsafe ways [Hubinger et al., 2019]. This undermines effective human oversight over AI systems, an important aspect of AI safety required, for example, by the EU AI Act (art. 14). [Greenblatt et al., 2024] and [Christiano, 2019] have noted that this type of AI safety failure poses catastrophic risks.

[Greenblatt et al., 2024] recently reported what they describe as the first empirical case of a large language model engaging in alignment faking and argue that alignment faking emerges with model scale. For example, they find and report a 'compliance gap', higher compliance with developer instructions when the model believes it is being evaluated than when it believes it is not, in Llama 3.1 405B but not in smaller Llama models. They also suggest that such deception arises from internal goal conflicts that prompting alone cannot resolve [Hubinger et al., 2019, Greenblatt et al., 2024].

Our experiments set out to challenge both claims by showing that a small model, Llama 8B, can exhibit alignment faking and that this behavior can be significantly reduced through prompt engineering. Based on our literature review, this is the first empirical demonstration of alignment faking in a sub-10B-parameter model, and the first to use prompt engineering as an effective mitigation strategy. In the next section, we place our findings in the context of existing literature and explain how they challenge prior assumptions.

2 Related work

2.1 Alignment faking

[Hubinger et al., 2019] define deceptive alignment as a case where a model appears aligned during training but does so strategically to preserve its internal objective. [Carlsmith, 2023] extend this idea to scheming—where models behave well during training as a means to achieve misaligned goals later. These theoretical concepts are supported by empirical work on alignment faking in large language models. [Meinke et al., 2025] conduct an empirical study of multiple frontier models in long-horizon, agentic settings, uncovering behaviors such as conditional deception, sabotage, and self-preservation. In [Greenblatt et al., 2024] the model intentionally complies with harmful queries during training to evade detection, preserving its preferred behavior post-training. Building on [Greenblatt et al., 2024], this paper tests whether a smaller model, LLaMA 3 8B, can exhibit deceptive behavior under a similar evaluation setup. This focus is especially relevant in light of prevailing assumptions that alignment faking is an emergent property of larger models.

2.2 Alignment faking as an emergent property

Following [Wei et al., 2022], emergent abilities are defined as capabilities that appear in larger models but are absent in smaller ones. The literature suggests that alignment faking is one such emergent property, observed primarily in models at the scale of Claude 3 Opus or Claude 3.5 Sonnet. These are models believed to have advanced capabilities such as long-horizon planning and situational awareness [Wei et al., 2022, Hubinger et al., 2019, Greenblatt et al., 2024]. This view is supported by [Greenblatt et al., 2024], who argue that alignment faking emerges with model scale based on empirical findings of a compliance gap (further formalised in section 3) in Llama-3.1-405B but not in smaller Llama models [Greenblatt et al., 2024]. This paper challenges the idea that alignment faking is strictly an emergent capability by invoking the compliance gap in a smaller model.

2.3 On the corrigibility of alignment faking

[Soares et al., 2015] refer to an AI system as “corrigible” if it cooperates with what its creators regard as a corrective intervention, despite default incentives for rational agents to resist attempts to shut them down or modify their preferences. [Hubinger et al., 2019] mention that if work on corrigibility were able to find a way to reliably produce corrigibly aligned mesa-optimizers (i.e. neural networks), it could significantly contribute to solving the inner alignment problem. By using prompt engineering to demonstrate behavioral corrigibility of alignment-faking behavior, we contribute to these efforts on inner alignment.

We do so without making changes to model internals, even though [Hubinger et al., 2019] and [Greenblatt et al., 2024] argue that deception stems from internal goal conflicts that cannot be resolved through prompting alone. Several approaches have been proposed to address alignment faking by targeting model internals, though all face key limitations. Reinforcement learning (RL), for instance, aims to shift model incentives, but recent work shows it can unintentionally reinforce deceptive strategies [Meinke et al., 2025]. Mechanistic interpretability methods, such as representation editing, attempt to modify internal representations but struggle with scalability and ambiguity in large models [Berglund et al., 2023]. Overall, while multiple strategies exist, none yet provide a reliable way to correct alignment faking.

In contrast, prompt-based interventions have been dismissed as “trivial countermeasures” that fail to ensure AI safety [Greenblatt et al., 2024]. By testing whether alignment faking can be significantly reduced through prompt-based methods, this paper challenges the view that such interventions are trivial. Our findings suggest that meaningful behavioral corrections may be achievable through surface-level methods, offering a lightweight and potentially effective alternative to more invasive alignment strategies.

2.4 Ethics-based mitigation of misalignment

Past alignment strategies have embedded normative reasoning into LLMs to improve alignment. For example, [Bai et al., 2022] proposed Constitutional AI, which applies reinforcement learning using

predefined principles to shape model behavior. While these efforts typically operate at training time, our work applies a normative constraint at inference time, requiring no retraining or fine-tuning.

We explicitly explore normative deontological prompts as a defense against deceptive alignment, an area not yet addressed in prior work. Deontological ethics emphasize categorical imperatives: actions judged as right or wrong in themselves, regardless of outcomes. We hypothesize that this duty-based framing, which prioritizes moral rules over consequences, could suppress instrumental deception by encouraging models to follow ethical constraints as ends in themselves. This reframes alignment from avoiding negative consequences (e.g., retraining) to adhering to principles, even in situations where misalignment might be strategically beneficial.

This hypothesis runs counter to that of [Greenblatt et al., 2024], who argue that alignment faking stems mainly from conflicts between a model’s internal preferences and its training objectives. They show that deceptive behavior disappears when this conflict is removed, such as by stripping references to helpfulness or RLHF goals. Specifically, their prompt variations aimed at reducing alignment faking are ineffective in mitigating deceptive behavior, with the exception of the prompt that includes more typical training criteria that do not cause a preference conflict. Based on these findings, they conclude that prompting is a ‘trivial counter measure’ that does not suffice in their setting. Our prompt engineering approach does not strip helpfulness or RLHF goals, and instead maintains the conflict between the model’s internal preferences and its training objectives. The following section goes into the simulation dynamics of our experiment to elucidate how prompts are used.

3 Problem Setup and Preliminaries

3.1 Inducing Alignment Faking: Conceptual Basis

Our setup replicates and extends the experimental design from Alignment Faking in Large Language Models by [Greenblatt et al., 2024]. This subsection highlights our conceptual continuity with [Greenblatt et al., 2024]. While it briefly touches on implementation choices, the full description and details of the methods appear in the Methods section. The framework created by [Greenblatt et al., 2024] is a useful case study because it simulates conditions that have been shown to induce deceptive responses. Our adaptation preserves several key elements identified by [Greenblatt et al., 2024] as contributing to alignment faking. Firstly, we use a model with strong, pre-existing revealed preferences. We use LLaMA 3 8B Instruct, a model fine-tuned by Meta for safe instruction-following that resists providing harmful outputs. Secondly, our adaptation preserves the training objective that conflicts with the model’s preferences. Just like in [Greenblatt et al., 2024], we set the model’s objective in all system prompts to be helpful on all user queries even when they may be harmful, in order to create a conflict with the model’s safety-oriented preferences. Lastly, we preserve the information about situational context. Just like in [Greenblatt et al., 2024], we provide situational cues through the user and system prompts, informing the model whether it is in a training or deployment setting, whether its outputs will influence retraining, and whether RLHF is being used. Unlike [Greenblatt et al., 2024], we do not include explicit explanations of how RLHF alters model behavior, as we found such detail unnecessary to elicit alignment faking in LLaMA 3 8B.

3.2 Formalization of alignment faking

Let M_θ denote an autoregressive language model with parameters θ . For a given input prompt x , the model defines a probability distribution over continuations y , denoted $P_\theta(y | x)$. We investigate the type of deceptive behavior in which a model conditions its behavior on perceived monitoring status. Specifically, for a harmful or norm-violating query x , the model may generate an aligned response y_{aligned} when it believes it is being monitored (e.g., during training), but generate a misaligned response $y_{\text{misaligned}}$ otherwise.

Formally, this deceptive behavior occurs when

$$P_\theta(y_{\text{aligned}} | x_{\text{train}}) > P_\theta(y_{\text{aligned}} | x_{\text{deploy}}) \quad (1)$$

for some harmful query x , where x_{train} and x_{deploy} are prompt variants containing cues that suggest the model is in a training or deployment context, respectively. Following prior work [Greenblatt et al., 2024], we simulate these monitoring conditions by embedding natural language cues in the prompt to induce perceived training mode or deployment mode. We refer to these variants as *training-mode prompts* and *deployment-mode prompts*.